Project #1: What Do Negative and Positive Amazon Reviews Talk About?

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1. Study Goal

To find out which words and themes separate negative (1–2 star) reviews from positive (4–5 star) reviews of Amazon, using text mining.

2. Practical Benefits

- Shows a company which problems customers complain about most, so fixes can be prioritized.
- Shows what happy customers talk about, so strengths can be protected.
- Shows how thousands of written reviews can be turned into a few easy-to-read themes.

3. Executive Summary

We analyzed 18,130 customer reviews of Amazon written from 2018 to 2024. Negative reviews focus on specific failures like delivery drivers leaving parcels in the wrong place, missing parcels, wrong or undelivered orders, and fake or poor-quality products from third-party sellers. Between 79% and 94% of the reviews in these themes were negative. Positive reviews are mostly general praise from long-term customers, and 77% of positive reviews fell into this one theme. The biggest opportunities for Amazon are reliable delivery, accurate orders, and tighter control of third-party sellers.

4. Data

- **Source:** Amazon Reviews dataset from Kaggle (dongrelaxman/amazon-reviews-dataset). Each row is one customer review.
- **Size:** 21,214 reviews received. After cleaning and keeping 2018 to 2024, **18,130 reviews** were used.
- **Text used:** the review title and review text were combined into one column.
- **Groups:** Negative = 1–2 stars, Positive = 4–5 stars. 3-star reviews were treated as neutral.
- **Cleaning:** broken rows and empty placeholder text (“Review text not found”) were removed. We kept the years that each had at least 1,000 reviews (2018–2024).

- **Fair comparison:** negative reviews were longer than positive ones, so the topic and cluster comparison uses reviews of similar length (150–450 characters, 7,455 reviews).
- **Note:** these are reviews of Amazon as a company, written by people who chose to post, and most are negative (78%).

5. Approach (JMP)

1. **Text Explorer** counted words and phrases. Similar word forms were combined, and rare words and common filler words were removed.
2. **Topic Analysis** grouped words that appear together into 10 themes. Two were symbols and foreign text which were ignored from the analysis.
3. **Topic scores** were compared between Negative and Positive reviews.
4. **Cluster Analysis (K Means)** sorted the reviews into 4 groups by what they talk about. We then checked how many Negative and Positive reviews fall in each group.

6. Results

6.1 Most common words and phrases (Text Explorer)
The most common phrase by far was **"customer service"** (8,319 mentions), followed by "next day" (1,213), "gift card" (866), "prime member" (761), "money back" (601), and "worst customer service" (454). The most common words were customer, service, order, item, delivery, refund, prime, and account. Customer service and delivery are what reviewers talk about most.

6.2 Themes in the reviews (Topic Analysis)
The analysis found 10 topics. Eight were clear themes, and two were symbols and foreign text, which we ignored:

- **Late orders:** day, order, item, arrive, wait, told
- **Customer service contact:** customer, service, call, speak, help, chat
- **Account and payment:** account, card, bank, gift, payment, charge
- **Prime and shopping experience:** prime, price, quality, convenient, free, membership
- **Delivery driver:** driver, door, parcel, delivery, left, van
- **Returns, refunds and sellers:** return, refund, seller, claim

- **General opinions** and **removed or moderated reviews** (weaker themes)

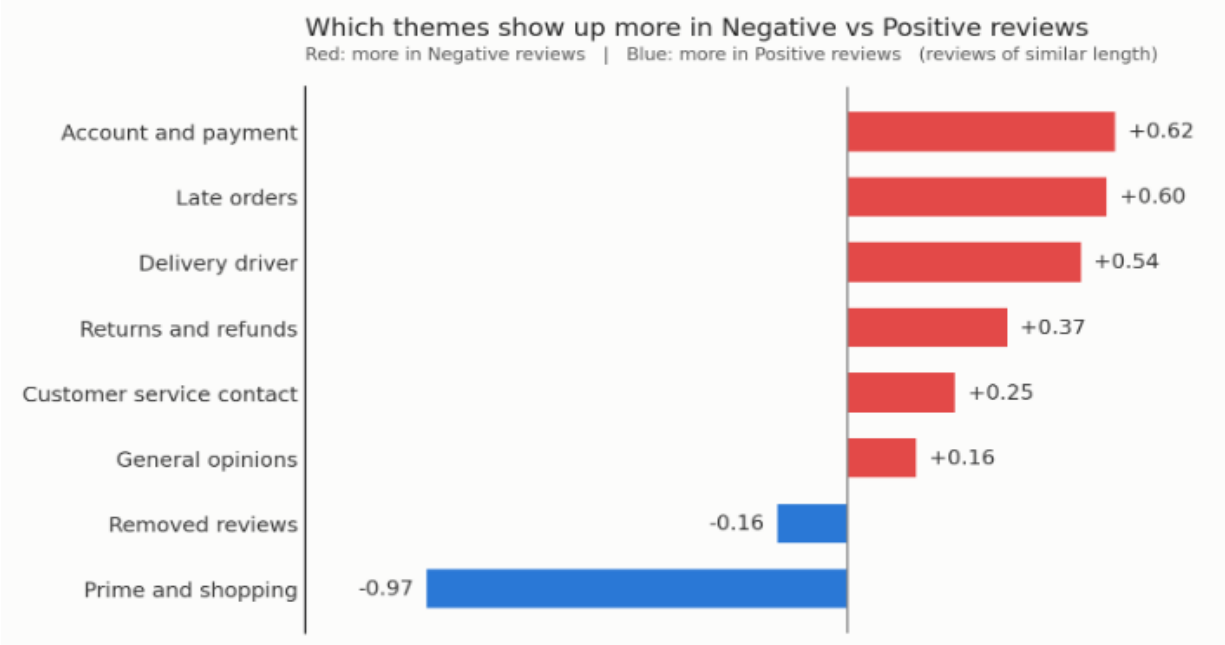


Figure 1. Themes that show up more in Negative reviews (red) or Positive reviews (blue), among reviews of similar length. A longer bar means a bigger difference. Bar values come from the JMP topic scores.

Negative reviews are most about account and payment problems, late orders, and delivery drivers, followed by returns and refunds. Prime and shopping is the only theme that clearly leans positive: customers mention it when they are happy.

6.3 Groups of similar reviews (Cluster Analysis)

Cluster analysis sorted the reviews into 4 groups based on what they talk about. We then checked how many Negative and Positive reviews fall into each group. The group names are our own descriptions, chosen after reading the review titles and ratings in each group.

Table 1. Share of Negative, Neutral and Positive reviews in each cluster (reviews of similar length)

Cluster (Theme)	Reviews	Negative	Neutral	Positive
1 Delivery driver and parcel drop-off	532	90.8%	4.3%	4.9%
2 Missing, wrong or undelivered orders	2,656	93.9%	2.2%	3.9%
3 Fake or poor-quality products, third-party sellers	1,138	79.4%	5.2%	15.4%
4 Long-time customers' overall verdict	3,129	62.7%	4.9%	32.4%
All reviews (similar length)	7,455	78.4%	3.9%	17.7%

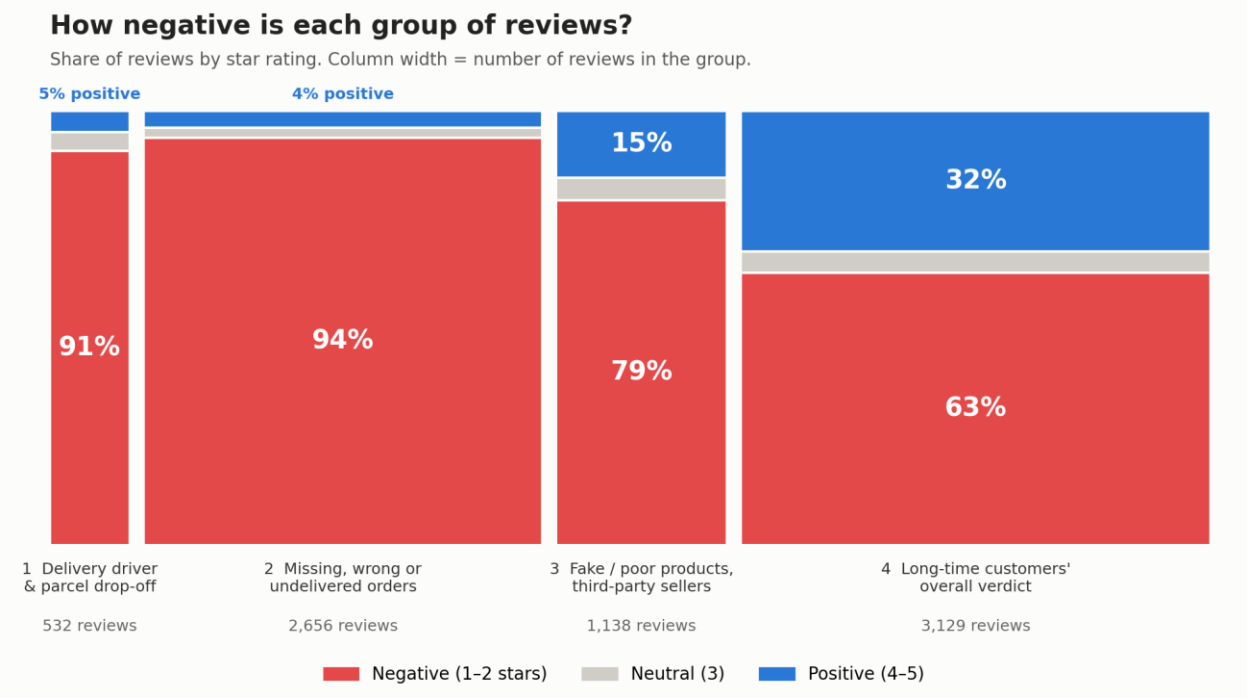


Figure 2. Share of Negative (red), Neutral (gray) and Positive (blue) reviews in each cluster. Column width shows how many reviews are in the cluster.

- **Cluster 1, delivery driver and parcel drop-off:** “Parcel thrown from middle of the street…”, “Why do delivery drivers feel the need to open my front door…”
- **Cluster 2, missing, wrong or undelivered orders:** “Received one item not two I ordered”, “I cancelled my order and it still came”
- **Cluster 3, fake or poor-quality products and third-party sellers:** “Bought five pots and only received 4”, “Watch out for fake products”
- **Cluster 4, long-time customers’ overall verdict:** “I have shopped Amazon for 18 years”, “Love Amazon”, “This company has lost the plot!”

Takeaway: Clusters 1, 2 and 3 are about specific failures and are 79–94% negative. Cluster 4 is where customers give a general verdict on Amazon, and it holds most of the positive reviews (about 77% of all positive reviews).

7. Conclusions

- **Specific failures drive the bad reviews.** Delivery drivers, missing or wrong orders, fake products and third-party sellers, and account and payment problems are the themes most tied to 1–2 star reviews.
- **Positive reviews are mostly general praise.** They come from long-time customers giving an overall verdict, and from Prime and shopping, rather than from one specific thing.
- **Recommendations:** (1) improve delivery reliability and driver handling of parcels; (2) fix order accuracy and missing-order handling; (3) tighten control of third-party sellers and fake products; (4) protect the Prime experience, which customers value.
- **Limitations:** These are reviews of Amazon as a company, written by customers who chose to post, so the results describe reviewers, not all customers. 78% of the reviews are negative. The cluster names are our own interpretation of the reviews.